Classification Tree Homework Solution

1. ***ID3 Tree Construction***:
    a. Dataset:

ID	Color	Shape	Size	Class
1	Red	Square	Big	+
2	Blue	Square	Big	+
3	Red	Round	Small	-
4	Green	Square	Small	-
5	Red	Round	Big	+
6	Green	Round	Big	-

    b. Initial entropy of dataset:
        i. 3 positives and 3 negatives:
            1. $$E(t) = -\left(\frac{3}{6}\log_2\frac{3}{6} + \frac{3}{6}\log_2\frac{3}{6}\right) = -2 \cdot (0.5 \cdot \log_2 0.5) = 1$$
    c. Information Gain:
        i. Color:
            1. Red = [+, -, +] = Entropy = 0.918
            2. Blue = [+] = Entropy = 0
            3. Green = [-,-] = Entropy = 0
            4. Weighted Average = $\left(\frac{3}{6}\right)(0.918) + \left(\frac{1}{6}\right)(0) + \left(\frac{2}{6}\right)(0) =$ 0.459
            5. **Gain(Color) = 1 -0.459 = 0.541**
        ii. Shape:
            1. Square = [+, +, -] = Entropy = 0.918
            2. Round = [-,+,-] = Entropy = 0.918
            3. Weighted Average = $\left(\frac{3}{6}\right)(0.918) + \left(\frac{3}{6}\right)(0.918) =$ 0.918
            4. **Gain(Shape) = 1 - 0.918 = 0.082**
        iii. Size:
            1. Big = [+, +, +, -] = Entropy = 0.811
            2. Small = [-,-] = Entropy = 0
            3. Weighted Average = $\left(\frac{4}{6}\right)(0.811) + \left(\frac{2}{6}\right)(0) =$ 0.54
            4. **Gain(Shape) = 1 - 0.54 = 0.46**
    d. ***Therefore, Color has the highest gain = split on Color first*****:**

Final Decision Tree:

[Color]

Red | Blue | Green

[?, ?, ?] | + | -

i. For Red (instances 1,3,5 = +, -, +):
   1. Split again = try shape
      a. Square = [+]
      b. Round = [-, +] = still mixed = try Size
         i. Big = [+]
         ii. Small = [-]

ii. Rules sample:
   1. If color = Blue -> class = +
   2. If color = Green -> class = -
   3. If color = Red and Shape = Square -> class = +
   4. If color = Red and Shape = Round and Size = Big -> class = +
   5. If color = Red and Shape = Round and Size = Small -> class = -

2. ***Impact of Adding New Attribute:***
   a. For example, let's say we add a new attribute called "Pattern of Shirt" with values: "Checked", "Striped", and "Plain".
   b. If this attribute has high information gain, it could become the new root of the decision tree. This might:
      i. **Restructure the entire tree** if the new attribute splits data more effectively, **improve classification accuracy** especially if it resolves ambiguity in the original dataset, or **change rule priorities** as new branches may be created with more refined splits.
   c. If a data scientist **misses this key attribute**, they could:
      i. Deliver **misleading insights** to decision-makers, cause the company to make **costly production errors, or** lose business opportunities or cause **strategic missteps**—especially if the missed attribute is **highly predictive**.
3. Summary:
   a. In this report, we built a decision tree using the ID3 algorithm, guided by entropy and information gain, to determine the most effective attribute for classification. After analyzing the dataset, Color emerged as the most decisive factor in predicting the outcome, heavily influencing how the data was split. However, the inclusion of a new attribute such as "Pattern of Shirt" could dramatically shift this outcome by introducing a more informative split, potentially altering the tree's structure and improving classification accuracy. This reveals how a single overlooked feature can change the predictive power of a model. An analyst's ability to identify and incorporate such variables can be the difference between a mediocre and a highly effective model.